

SUSHRUTH VELDI

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SUMMARY

Software Engineer with 2+ years of experience building scalable backend systems, APIs, and cloud-native applications. Skilled in Python (FastAPI, Django, Flask) and Java (Spring Boot) microservices, with additional experience in React.js and Node.js for full-stack solutions. Strong background in cloud deployment (AWS, GCP, Azure), DevOps (Docker, Kubernetes, CI/CD), and data engineering pipelines, delivering reliable, high-performance software across enterprise environments.

EXPERIENCE

TGS-NOPEC OIL & GAS

Houston, Texas

Software Developer

May 2025- Present

- Developed a React application with a FastAPI backend to validate SEGY and MDIO seismic data by generating trace maps, extracting spatial footprints, and performing geometric comparisons with reference data.
- Optimized performance using Ray for parallel processing (60% faster trace reading), containerized with Docker, and added interactive visualizations with GeoPandas, OpenCV, and Plotly.
- Automated validation workflows, leveraging LLMs for metadata extraction and structured storage, ultimately reducing 400+ man-hours annually and saving significant time and costs.
- Worked on extracting trace headers from seismic data, designing a schema using Pydantic, and storing the results in a structured format for easier access and retrieval.

The George Washington University

Washington, DC

Full Stack Software Engineer - Graduate Assistant

Sep 2024-Jan 2025

- Developed a full-stack academic analytics dashboard (React, TypeScript, Node.js, MongoDB) for 200+ students, automating grading insights and lab performance trends; reduced manual grading effort by 60% with real-time visualizations, role-based access, and secure APIs (OAuth2, JWT, RBAC)
- Containerized and deployed with Docker on Google Cloud Run, enabling reliable, scalable access for instructors and TAs during peak hours; explored LLM integration (Gemini, OpenAI) to auto-summarize feedback and prototype AI-driven assistants that flagged performance gaps using NLP pipelines.

Larsen & Turbo Infotech

Hyderabad

Software Engineer

Aug 2021 – Sep 2022

- Built backend microservices using Python (FastAPI, Flask) and AWS (Lambda, S3, DynamoDB, EC2), improving onboarding throughput by 30% and ensuring reliability.
- Developed and deployed RESTful APIs with OAuth2 and JWT authentication, cutting response latency by 50% and strengthening access control.
- Automated CI/CD pipelines with Jenkins, AWS CodePipeline, and Kubernetes (Helm), enabling zero-downtime deployments and reducing release time by 70%.
- Collaborated with PMs, QA, and engineers in Agile sprints to deliver features aligned with business goals and compliance standards.

Verzeo

Bengaluru

Software Engineering Intern

May 2021 - July 2021

- Contributed to a cloud-based ERP system using Java (Spring Boot) for inventory, HR, and finance modules; improved efficiency by streamlining backend services and integrating REST/GraphQL APIs
- Assisted in deploying microservices to cloud platforms (Azure, AWS) with Docker and Kubernetes, implementing secure authentication (JWT, Azure AD) and supporting Agile-based delivery for scalable solutions.

TECHNICAL SKILLS

- Programming & Scripting Languages:** Python, R, Java, SQL, C, C++, HTML, CSS, JavaScript, Bash, Kotlin
- Databases :** MongoDB, MySQL, PostgreSQL, SQL Server, SQLAlchemy
- Frameworks & Libraries:** Fastapi, React, Flask, Django, Spring Boot, React.js, Node.js, NumPy, Pandas, Scikit-learn, TensorFlow, OpenCV
- Data & Analytics Tools:** Tableau, Power BI, Excel, Apache Spark, Hadoop, Datawrapper
- Cloud & DevOps:** AWS, GCP, Azure, Docker, Kubernetes, Jenkins, Azure DevOps, Git, Linux

EDUCATION

The George Washington University

Washington, DC

Master of Science in Data Science

January 2025

Jawaharlal Nehru Technological University

Hyderabad, India

Bachelor of Technology in Computer Science

August 2022

CERTIFICATIONS

AWS Certified Cloud Practitioner – Amazon Web Services (AWS) | Issued: Oct 2024